

eLearning

Computing and Mathematics

May, 2026



Short Title:	eLearning	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Media Production	
Author	CFITZPATRICK	
Credits	5	Level: Advanced

Description of Module / Aims

This module is concerned with the authoring of instructionally sound, accessible eLearning for different platforms. This can, for example, be a single part of a course, or all of a course whether it is in an academic environment, part of mandatory business training or a full distance learning course.

Programmes

COMP -0636	BSc (Hons) in Creative Computing (WD_KCRCO_B)	4 / 8 / E
COMP -0636	BSc (Hons) in Multimedia Applications Development (WD_KMULM_B)	4 / 2 / E

Indicative Content

- Design and Authoring of eLearning
- Needs and Task Analysis
- Organising Instruction
- Learning Environments
- Using Standards and Guidelines
- Creating, Testing and Validation of Accessible Web Sites

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate and critique the suitability of eLearning Authoring Tools.
2. Design organised instruction and develop learning environments using Instructional Design Guidelines.
3. Prepare needs and task analysis.
4. Substantiate the creation, testing and validation of accessible user interfaces using guidelines and standards.
5. Develop a computer based, accessible learning environment.

Learning and Teaching Methods

- This module will be delivered using 2 hours of computer-based Lectures, along with 2 hour of computer-based Practicals each week.
- The Lectures will introduce new concepts and theory relating to eLearning. The practicals will allow the students to design and develop eLearning content using industry standard eLearning software.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	20%	1
Assignment	30%	2,3
Assignment	50%	4,5

Assessment Criteria

<40%: Unable to interpret and describe key concepts of the eLearning.

40%-49%: Be able to interpret and describe key concepts of the eLearning.

50%-59%: Ability to discuss key concepts of eLearning and ability to discover and integrate related knowledge in other knowledge domains.

60%-69%: Be able to solve problems within eLearning by experimenting with the appropriate skills and tools.

70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Clark, R. and R. Mayer. *e-Learning and the Science of Instruction: Proven Guidelines for Consumers and Designers of Multimedia Learning*. 3rd ed.. United States of America: Pfeiffer, 2011.

Supplementary Material

- Horton, W. *E-Learning by Design*. 2nd ed.. United States of America: Pfeiffer, 2011.

Requested Resources

- COMPUTER LAB: Multimedia Lab